Q1.

Diagram **A** represents a gas in a container.
The gas can be compressed by moving the piston to the right.

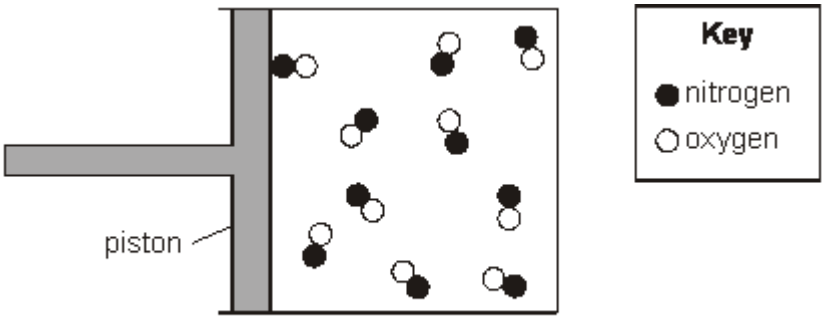


diagram A

(a) (i) How can you tell that the substance in the container is a gas?

.....

.....

1 mark

(ii) How can you tell from the diagram that the gas is pure?

.....

.....

1 mark

(b) The piston is moved to the right as shown in diagram **B**.

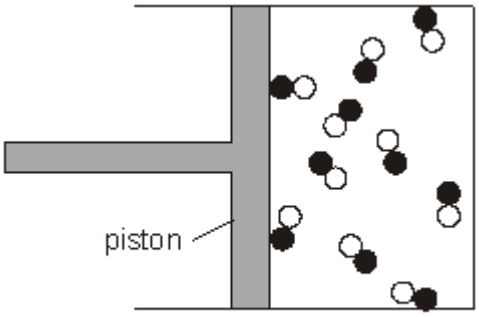


diagram B

How can you tell, from diagram **B**, that the pressure of the gas has increased?

.....

.....

1 mark

(c) Diagram **C** shows what happened to the molecules after the gas was compressed more.

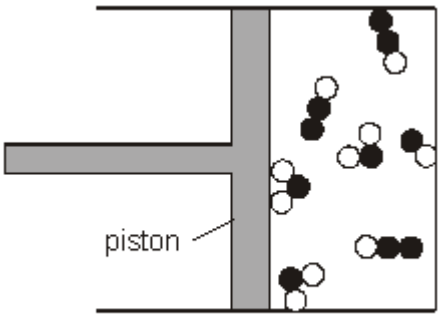


diagram C

(i) How can you tell that a chemical reaction happened when the gas was compressed?

.....

.....

1 mark

(ii) The mass of the gas in both diagrams **B** and **C** was 0.3 g.

Why did the mass of the gas **not** change when it was compressed?

.....

.....

1 mark

(iii) Complete the table below with the correct chemical formula of each substance. Use the key to help you.

substance	formula
●○	
●●○	
	

Key
● nitrogen
○ oxygen

1 mark

(iv) What is the **name** of the substance represented by the symbol ●○?

.....

1 mark
maximum 7 marks

Q1.

(a) (i) • (molecules) are far apart **or** not touching each other

accept 'only gases can be compressed'

'the gas can be compressed' is insufficient as it is given in the question

accept 'they are randomly arranged'

1 (L7)

(ii) • there is only one type of molecule **or** compound **or** substance

accept 'there is one type of particle'

*do **not** accept 'there is only one type of atom **or** element'*

1 (L7)

(b) any **one** from

• the space **or** distance between the molecules **or** particles is smaller

accept 'the volume is less'

accept 'atoms' for 'particles'

• the particles **or** they are closer together

• more particles are touching the sides

accept 'particles hit the sides more often'

'the particles are hitting the sides' is insufficient

'if the gas is compressed the pressure rises' is insufficient

1 (L7)

(c) (i) any **one** from

• new **or** different compounds have formed

accept 'they are now joined in threes'

*accept 'new combinations of particles **or** atoms'*

• there is more than one compound

accept 'the compounds are different'

accept 'there is no longer a pure substance'

1 (L7)

(ii) any **one** from

• the same number of atoms are present

accept 'mass is conserved'

'the mass stays the same' is insufficient

• nothing has been added to **or** lost

'the same atoms are present' is insufficient

'nothing changed' is insufficient

'the amount of gas stays the same' is insufficient

1 (L7)

(iii) •

	NO	*accept 'ON'*
	N_2O	*accept 'ON_2'*
	NO_2	*accept 'O_2N'*

***all three** answers are required for the mark*

1 (L7)

(iv) • nitrogen oxide

accept 'nitrogen monoxide'

accept 'nitric oxide'

1 (L7)

[7]